

SHANKAR SHARMA

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Education

PhD, Climate Change Research Centre (CCRC), University of New South Wales, Sydney
Urban Climate | 2023–present

Masters, Institute of Tibetan Plateau Research, UCAS, Beijing, China
Atmospheric Physics and Atmospheric Environment | 2016–2020

Bachelors, Tri-Chandra Multiple Campus, Ghantaghar, Kathmandu, Nepal
Hydrology and Meteorology | 2011–2014

Professional Experience

Research Associate and Tutor
Central Department of Hydrology and Meteorology, Tribhuvan University, Nepal | 2020–2023

Climate Analyst –Consultant
Worldwide Fund for Nature | 2020

Junior Hydrologist
RECHAM Consult Pvt. Ltd. | 2014 – 2016

Awards

- University International Postgraduate Award (UIPA), UNSW Sydney (2023)
- Incentive Award (2021), Nepal Academy of Science and Technology (NAST), for research on precipitation variability and MERRA-2 precipitation evaluation
- UCAS Master's Scholarship Award (2016), China

Conference and Trainings

- **Oral Presentation**, Australian Meteorological and Oceanographic Society (2025 and 2026)
- **Oral Online Presentation**, International Conference on Urban Climate (ICUC), 2025
- **Poster Presentation**, ARC Centre of Excellence for Climate Extremes / 21st Century Weather and Climate Annual Workshop 2024 (Sydney) and 2025 (Launceston).
- International Workshop on ICTP Regional Climate Model, Dibrugarh University (22–28 Feb 2021)
- Climate Change Indices Analysis Training using CDAT and CORDEX datasets (21–24 Jun 2022)
- Annual Winter School 2024: Weather and Climate Interactions, Monash University, Melbourne (Jun 2024)

Academic Service and Leadership

Coordinator, CCRC Land Group Meeting, UNSW Sydney
Bi-weekly organization of research discussions, presentation scheduling, and meeting facilitation.

Technical Skills

Python; WRF-ARW modelling; MATLAB; ArcGIS

Publications

⁸ <https://scholar.google.com/citations?user=gjTxjbgAAAAJ&hl=en>

2025

Sharma, S., Evans, J.P., Pitman, A., Nazarian, N., Lipson, M.J., Lopez-Bravo, C. (2025). Mapping urban dynamics in Greater Sydney—A scalable multi-decadal local climate zone classification approach. *PLOS Climate* 4(8), e0000677. <https://doi.org/10.1371/journal.pclm.0000677>

Sharma, S., Dahal, P., Marahatta, S., Shrestha, A., Panday, V.P., Pokharel, B. Historical trend and future projection of extreme precipitation in Nepal from high-resolution CMIP6 data. In *Severe Storms: Anatomy, Early Warning Systems and Aftermath in Changing Climate Scenarios* (pp. 607–625). Springer Nature Singapore (2025).

2024

Khadka, N., Chen, X., Liu, W., Gouli, M. R., Zhang, C., Shrestha, B., & **Sharma, S.** (2024). Glacial lake outburst floods threaten China–Nepal connectivity: Synergistic study of remote sensing, GIS, and hydrodynamic modeling with regional implications. *Science of The Total Environment*, 948, 174701. <https://doi.org/10.1016/j.scitotenv.2024.174701>

2023

Sharma, S., Hamal, K., Pokharel, B., Fosu, B., Wang, S.-Y. S., Gillies, R. R., Aryal, D., Shrestha, A., Marahatta, S., & Hussain, A. (2023). Atypical forcing embedded in typical forcing leading to the extreme summer 2020 precipitation in Nepal. *Climate Dynamics*, 61(7), 3845–3856. <https://doi.org/10.1007/s00382-023-06777-9>

Pokharel, B., **Sharma, S.**, Hamal, K., Stuienvolt-Allen, J., Wang, S.-Y. S., LaPlante, M., Gillies, R. R., Khanal, S., Wehner, M., & Rhoades, A. (2023). Amplified drought trends in Nepal increase the potential for Himalayan wildfires. *Climatic Change*, 176(2), 17. <https://doi.org/10.1007/s10584-023-03495-3>

Dawadi, B., **Sharma, S.**, & Reynard, E. (2023). Climatology, variability, and trend of the winter precipitation over Nepal. *Earth Systems and Environment*, 7(2), 381–391. <https://doi.org/10.1007/s41748-023-00338-0>

Khadka, N., Chen, X., Sharma, S., Shrestha, B. (2023). Climate change and its impacts on glaciers and glacial lakes in Nepal Himalayas. *Reg Environ Change* 23, 143 (2023). <https://doi.org/10.1007/s10113-023-02142-y>

Yang, H., Yang, K., Qin, J., Ding, B., Jiang, Y., Chen, Y., Zhou, X., Wang, Y., & **Sharma, S.** (2023). Adaptive statistical spatial downscaling of precipitation supported by high-resolution atmospheric simulation data for mountainous areas of Nepal. *Journal of Meteorological Research*, 37(4), 508–520. <https://doi.org/10.1007/s13351-023-3017-6>

2022

Talchabhadel, R., **Sharma, S.**, Khadka, N., Hamal, K., Karki, S., & Thapa, B. R. (2022). An outlook on the applicability of satellite precipitation products for monitoring extreme precipitation events in Nepal Himalaya. *Weather*, 77(5). <https://doi.org/10.1002/wea.4143>

Shrestha, B., Zhang, L., **Sharma, S.**, Shrestha, S., & Khadka, N. (2022). Effects on ecosystem services value due to land use and land cover change (1990–2020) in the transboundary Karnali River Basin, Central Himalayas. *SN Applied Sciences*, 4(5), 137. <https://doi.org/10.1007/s42452-022-05022-y>

Hamal, K., Ghimire, S. K., Khadka, A., Dawadi, B., & **Sharma, S.** (2022). Interannual variability of spring fire in southern Nepal. *Atmospheric Science Letters*, 23(9), e1096. <https://doi.org/10.1002/asl.1096>

Khadka, N., Ghimire, S. K., **Sharma, S.**, & Hamal, K. (2022). Rapidly expanding glacial lakes in Nepal Himalaya. *Jalawaayu*, 2(1), 45–55. <https://doi.org/10.3126/jalawaayu.v2i1.45393>

Pokharel, B., Hamal, K., Zhao, L., Aryal, D., Rajbhandari, D. P., Marahatta, S., & **Sharma, S.** (2022). Revisiting the 1993 extreme precipitation in central Nepal: Synoptic feature and numerical simulation. *Jalawaayu*, 2(1), 33–44. <https://doi.org/10.3126/jalawaayu.v2i1.45393>

Hussain, A., Cao, J., Ali, S., Ullah, W., Muhammad, S., Hussain, I., Abbas, H., Hamal, K., **Sharma, S.**, & Akhtar, M. (2022). Wavelet coherence of monsoon and large-scale climate variabilities with precipitation in Pakistan. *International Journal of Climatology*, 42(16), 9950–9966. <https://doi.org/10.1002/joc.7874>

2021

Sharma, S., Hamal, K., Khadka, N., Shrestha, D., Aryal, D., & Thakuri, S. (2021). Drought characteristics over Nepal Himalaya and their relationship with climatic indices. *Meteorological Applications*, 28(2), e1988. <https://doi.org/10.1002/met.1988>

Sharma, S., Khadka, N., Nepal, B., Ghimire, S. K., Luintel, N., & Hamal, K. (2021). Elevation dependency of precipitation over southern slope of central Himalaya. *Jalawaayu*, 1(1), 1–14. <https://doi.org/10.3126/jalawaayu.v1i1.36446>

Sharma, S., Hamal, K., Khadka, N., Ali, M., Subedi, M., Hussain, G., Dawadi, B. (2021). Projected drought conditions in the southern slope of Central Himalaya using CMIP6 models. *Earth System and Environment*, 5(3). <https://doi.org/10.1007/s41748-021-00254-1>

Shrestha, D., **Sharma, S.**, Talchabhadel, R., Deshar, R., Hamal, K., Khadka, N., Nakamura, K. (2021) Detection of spatial rainfall variation over the Andean region demonstrated by satellite-based observations. *Atmosphere*, 12(9):1204. <https://doi.org/10.3390/atmos12091204>

Chen, Y., **Sharma, S.**, Zhou, X., Yang, K., Li, X., Niu, X., Hu, X., & Khadka, N. (2021). Spatial performance of multiple reanalysis precipitation datasets on the southern slope of central Himalaya. *Atmospheric Research*, 250, 105365. <https://doi.org/10.1016/j.atmosres.2020.105365>

Hamal, K., **Sharma, S.**, Pokharel, B., Shrestha, D., Talchabhadel, R., & Shrestha, A. (2021). Changing pattern of drought in Nepal and associated atmospheric circulation. *Atmospheric Research*, 262, 105798. <https://doi.org/10.1016/j.atmosres.2021.105798>

Hamal, K., **Sharma, S.**, Talchabhadel, R., Ali, M., Dhital, YP., Xu, T., Dawadi, B. (2021). Trends of Diurnal Temperature Range over the Southern Slope of Central Himalaya: Retrospective and Prospective Evaluation. *Atmosphere*, 12, 1683. <https://doi.org/10.3390/atmos12121683>

Shrestha, D., **Sharma, S.**, Bhandari, S., & Deshar, R. (2021). Statistical downscaling and projection of future temperature and precipitation change in the Gandaki Basin. *Journal of Institute of Science and Technology*, 26(1), 16–27. <https://doi.org/10.3126/jist.v26i1.37810>

Nepal, B., Shrestha, D., **Sharma, S.**, Shrestha, M. S., & Aryal, D. (2021). Assessment of GPM-Era satellite products' ability to detect precipitation extremes over Nepal. *Atmosphere*, 12(2), 254. <https://doi.org/10.3390/atmos12020254>

Ali M.; Fayaz A., Zhu H., Ahmed M., Sigdel SR., Huang R.; **Sharma S.**, Liang E.; Hussain I., and Yaseen T. (2021). Dendrochronological Investigation of Selected Conifers from Karakoram-Himalaya, Northern Pakistan. *Pakistan Journal of Botany*, 53(3). [http://dx.doi.org/10.30848/PJB2021-3\(20\)](http://dx.doi.org/10.30848/PJB2021-3(20))

2020

Sharma, S., Khadka, N., Hamal, K., Shrestha, D., Talchabhadel, R., & Chen, Y. (2020). How accurately can satellite products (TMPA and IMERG) detect precipitation patterns, extremities, and drought across the Nepalese Himalaya? *Earth and Space Science*, 7(8), e2020EA001315. <https://doi.org/10.1029/2020EA001315>

Sharma, S., Chen, Y., Zhou, X., Yang, K., Li, X., Niu, X., Hu, X., & Khadka, N. (2020). Evaluation of GPM-Era satellite precipitation products on the southern slopes of the central Himalayas against rain gauge data. *Remote Sensing*, 12(11), 1836. <https://doi.org/10.3390/rs12111836>

Sharma, S., Khadka, N., Hamal, K., Baniya, B., Luintel, N., & Joshi, B. B. (2020). Spatial and temporal analysis of precipitation and its extremities in seven provinces of Nepal (2001–2016). *Applied Ecology and Environmental Sciences*, 8(2), 64–73. <https://doi.org/10.12691/aees-8-2-4>

Sharma, S., Hamal, K., Khadka, N., Joshi, B. B., & Baniya, B. (2020). Dominant pattern of year-to-year variability of summer precipitation in Nepal during 1987–2015. *Theoretical and Applied Climatology*, 142(3), 1071–1084. <https://doi.org/10.1007/s00704-020-03359-1>

Hamal, K., **Sharma, S.**, Baniya, B., Khadka, N., & Zhou, X. (2020). Inter-annual variability of winter precipitation over Nepal coupled with ocean-atmospheric patterns during 1987–2015. *Frontiers in Earth Science*, 8, 161. <https://doi.org/10.3389/feart.2020.00161>

Hamal, K., **Sharma, S.**, Khadka, N., Baniya, B., Ali, M., Shrestha, M. S., Xu, T., & Shrestha, D. (2020). Evaluation of MERRA-2 precipitation products using gauge observation in Nepal. *Hydrology*, 7(3), 40. <https://doi.org/10.3390/hydrology7030040>

Hamal, K., **Sharma, S.**, Khadka, N., Haile, G. G., Joshi, B. B., Xu, T., & Dawadi, B. (2020). Assessment of drought impacts on crop yields across Nepal during 1987–2017. *Meteorological Applications*, 27(5), e1950. <https://doi.org/10.1002/met.1950>

Yue, S., Yang, K., Lu, H., Chen, Y., **Sharma, S.**, Yang, X., & Shrestha, M. L. (2020). Distinct temperature changes between north and south sides of central-eastern Himalayas since the 1970s. *International Journal of Climatology*, 40(9), 4300–4308. <https://doi.org/10.1002/joc.6439>

Khadka, N., Ghimire, SK., Chen, X., Thakuri, S., Hamal, K., Shrestha, D., **Sharma, S.** (2020). Dynamics of maximum Snow Cover Area and Snow Line Altitude across Nepal using improved MODIS data, 2003–2018. *Journal of Institute of Science and Technology*, 25(2), 17–24. <https://doi.org/10.3126/jist.v25i2.33729>

Other Publications

Shankar Sharma and Binod Pokharel. Nepal under severe drought condition, (newspaper article, *The Himalayan Times*, 3rd March, 2021). Available at: <https://thehimalayantimes.com/opinion/nepal-under-severe-drought-condition-winter-crops-affected>.

Binod Pokharel and **Shankar Sharma**. Conditions and control measures of air pollution, (online article, *onlinekhabar*, 15th April, 2021). Available at: <https://www.onlinekhabar.com/2021/04/945818>.

Shankar Sharma and Binod Pokharel. Drought and fire threats, (newspaper article, *The Himalayan Times*, 25 October, 2021). Available at: <https://thehimalayantimes.com/blogs/drought-and-fire-threats>.

References

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| 1. | Prof. Jason P. Evans
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